# Predictive Modeling of Midday Water Potentials: Identifying Key Drivers of Tree Water Stress

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## Abstract

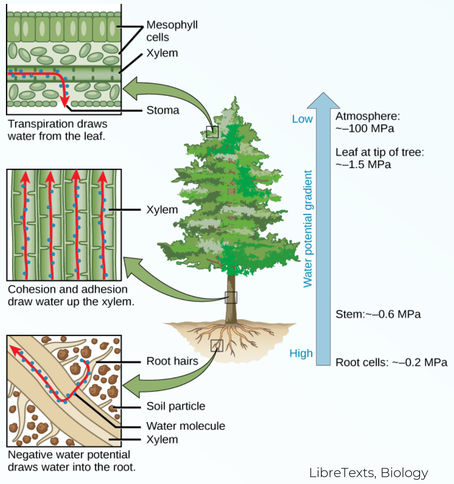
Understanding the factors influencing tree water stress is critical for predicting forest responses to changing climate. This study investigates primary drivers of midday water potential across trees at multiple field sites, integrating physiological traits and environmental conditions into a unified dataset. Results indicate that both environmental and physiological factors contribute to variation in midday water potential. With multiple linear regression and random forest statistical modeling, this study aims to better understand tree stress responses to environmental changes and improve predictions of behavior under future climate scenarios.

## What is water potential?

**Water potential**, measured in megapascals, indicates how easily water moves through a tree's vascular system and serves as a measure of plant water stress.

Predawn water potential is taken before sunrise, when the tree and surrounding soil are assumed to be in equilibrium. Midday water potential is measured during the hottest part of the day, when trees typically experience the most intense water stress due to high evaporative demand.

**Study Objectives:** Identify the main factors influencing midday water potential, with both environmental conditions and tree physiological traits. Evaluate the influence of including predawn water potential as a predictor by comparing models with and without this variable.

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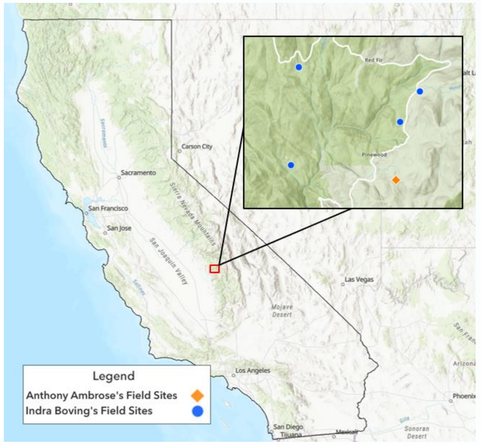
## Data and Methods

Midday water potential was modeled using Multiple Linear Regression, to estimate linear effects, and Random Forest, to capture nonlinear interactions and rank variable importance. Correlation heatmaps were used to explore relationships among variables and detect multicollinearity.

**Conifer Dataset 1** (Indra Boving, 2023): 240 observations from 60 trees across three conifer species: White Fir, Sugar Pine, and Incense Cedar.

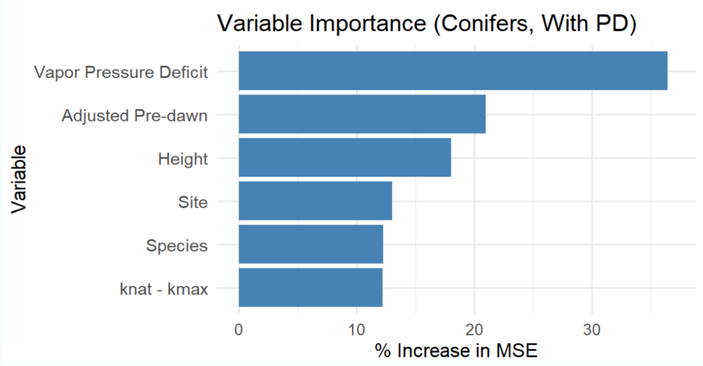
**SEGI Dataset 2** (Anthony Ambrose, 2015–2017): 431 observations from 81 unique Giant Sequoias.

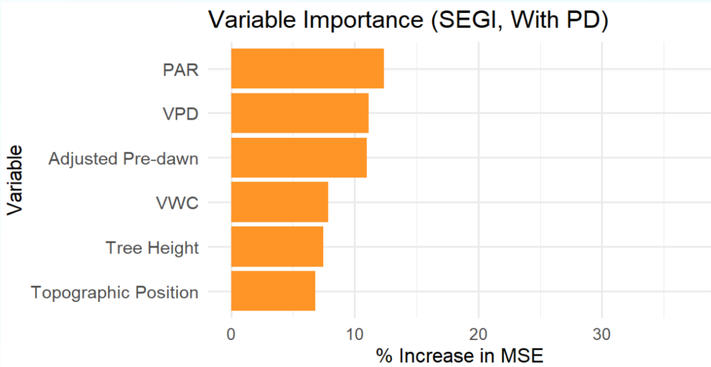
Both datasets include variables of tree size, water potentials, and environmental conditions, but the conifer dataset includes hydraulic trait data.




## Results

Effects of Atmospheric Dryness on Midday Water Potential is Modified by Plant Traits




**Key Predictors:**
Tree height, adjusted predawn water potential, and environmental variables like vapor pressure deficit (VPD) and photosynthetically active radiation (PAR) emerged as strong predictors of midday water stress.

**Model Performance:**
Conifers Dataset 1:
- Linear model: Explained ~50% of variation in MD
- Random forest: ~80% variance explained with PD; ~60% without

SEGI Dataset 2:
- Linear model: ~30% variance explained with PD; ~20% without
- Random forest: ~25% variance explained overall

Including predawn water potential improved model performance, particularly for conifers. Models were examined both with and without predawn water potential, given that it is a difficult measurement to obtain. This approach helped assess how much midday water potential could be explained in its absence, although including predawn water potential still generally improved model performance.

**Variable Importance:** %IncMSE plots from the random forest models rank predictors based on their contribution to model accuracy by measuring how much each variable impacts the model's performance when removed. They indicate VPD and pre-dawns as top predictors. In the conifer dataset, tree height was more influential, while in the SEGI dataset, PAR played a large role.

## Discussion

**Takeaways:** Model performance differed between datasets, possibly due to species-specific responses and data availability. In the conifer dataset, midday water potential was closely linked to predictors like tree height and predawn water potential. Results from the conifer dataset suggest that adding physiological data, such as hydraulic conductivity, could improve model accuracy in the SEGI dataset.

**Limitations:** $R^2$ values may not imply causation, as there may be other unmeasured variables such as root depth or canopy structure. Many environmental predictors are correlated, making it difficult to disentangle their individual effects. With only 75 observations in the conifer dataset and 155 in the SEGI dataset after removing missing values, the statistical models may have limited accuracy due to the relatively small sample size.

## Future Work

Plans include incorporating new data collected in 2024 into the existing models. The analysis will also be expanded to include seedlings to examine how environmental drivers differ between seedlings and mature trees and their implications for the survival and growth of young trees, as well as analyzing individual species within the conifer dataset to better understand species-specific responses. Additionally, alternative modeling approaches will be explored to improve predictive accuracy and generalizability.

## Acknowledgements & References

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Transport of Water and Solutes in Plants. 24 Dec. 2022, https://bio.libretexts.org/@go/page/97189.